

RESEARCH ARTICLE

Cell Culture and Tissue Engineering



Effect of inner diameter, filter length, and pore size on hollow fiber filter fouling during perfusion cell culture

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Abstract

As the need for higher volumetric productivity in biomanufacturing grows, biopharmaceutical companies are increasingly investing in a perfusion cell culture process, most commonly one that uses a hollow fiber filter as the cell retention device. A current challenge with using hollow fiber filters is fouling of the membrane, which reduces product sieving and can increase transmembrane pressure (TMP) past process limitations. In this work, the impact of hollow fiber filter geometries on product sieving and hydraulic membrane resistance profiles is evaluated in a tangential flow filtration (TFF) perfusion system. The hollow fibers tested had lengths ranging from 19.8 to 41.5 cm, inner diameters (IDs) ranging from 1.0 to 2.6 mm, and pore sizes of 0.2 or 0.65 μm . The results showed that the shortest hollow fibers experienced higher product sieving while larger IDs contributed to both higher product sieving and lower hydraulic membrane resistances, illustrating the impact of filter geometry on process performance. The results also showed 0.2 μm pore size filters maintain higher product sieving, but also higher membrane resistances compared to 0.65 μm pore size filters. This study highlights the need for optimized hollow fiber filter geometries to maximize use of the membrane area, which in turn can reduce production costs and increase scalability of the perfusion process.

KEYWORDS

continuous manufacturing, perfusion cell culture, product retention, tangential flow filtration, upstream bioprocess development

1 | INTRODUCTION

Biopharmaceutical companies are increasingly investing in perfusion cell culture processes over traditional fed-batch processes to meet higher market demands, produce more complex protein-based therapeutics, and realize the industry vision of a truly continuous biomanufacturing process. Perfusion processes can reach higher viable cell densities (VCDs), and as a result higher volumetric productivity, while using less manufacturing facility space than fed-batch processes. This is achieved

through a cell retention device that allows for the removal of spent media and product while fresh media is supplied to the cell culture.

A commonly used cell retention device is the hollow fiber filter set up in either an alternating tangential flow filtration (ATF) or tangential flow filtration (TFF) configuration. Despite their wide-spread use, hollow fibers still come with a number of challenges. One of the main challenges is fouling of the hollow fiber membrane, which reduces product sieving (product is retained in the bioreactor) and can increase the transmembrane pressure (TMP) past process limitations.

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Historically, ATF systems have been known to experience less fouling.^{1,2} However, the shift from peristaltic pumps to Levitronix's magnetically levitated centrifugal pump has improved the product sieving of TFF systems,³ in some cases to comparable levels of an ATF system,² because it imposes less mechanical stress on cells.⁴ Despite the higher product sieving seen with ATF systems, TFF systems are generally preferred in the industry regarding scalability concerns.⁵

Several papers have studied different filters^{6–8} or adjusted process parameters^{9,10} to reduce filter fouling and in turn maintain higher product sieving and lower transmembrane pressures. Wang et al. evaluated pore sizes ranging from 500 kDa to 0.45 μm with day 8 cell culture supernatant and found the 0.2 μm filter showed slightly higher product sieving than the other sizes.⁸ This result suggests there is an optimal size cut off that minimizes particle accumulation in the pores. Pinto et al. corroborated these results by showing the Asahi Kasei Microza 0.2 μm filter outperformed the equivalent 0.65 μm filter in terms of product sieving.⁶ In regards to TMP however, Stressmann et al. found that a 0.2 μm filter experienced a more severe TMP increase compared to the equivalent 0.45 μm filter at the same conditions.⁹ Pore sizes greater than 1 μm have also been assessed and show higher product sieving over 0.2 μm filters, however these pose a greater challenge to downstream processes.^{7,8} Studies have also assessed the effect of shear rate through the fiber: Stressmann et al. found that higher shear rates lead to higher end-of-run hydraulic membrane resistances⁹ and Pinto et al. saw sharper TMP increases at higher shear rates.⁶

In addition to experimental studies, several models, including computational¹¹ and empirical¹² models, have been developed to better understand and predict the fouling of hollow fiber filters. Radoniqi et al. developed a computational fluid dynamic model for fluid and pressure profiles in an ATF system that provided evidence for the presence of back-filtration,¹¹ also known as Starling flow, a phenomenon where local fluid flux is positive (entering the shell side) at the inlet of the filter but switches direction and becomes negative at the outlet (returning to the retentate side) due to the pressure gradient along the length of the fiber. Binabaji et al. modeled the same effect in hollow fiber filters for ultrafiltration of highly concentrated antibodies, also showing experimentally that higher axial flow rates, which increase the axial pressure drop, reduced the maximum achievable protein concentration due to an increase in back-filtration.¹³ In a subsequent paper, Binabaji et al. similarly showed shorter hollow fiber filters experienced lower axial pressure drops, leading to less back-filtration and higher maximum achievable antibody concentrations. Their data aligned with the theoretical axial pressure drop calculated using the Hagen-Poiseuille equation.¹⁴

This work investigates different hollow fiber filters to assess contributions of pore size, inner diameter (ID), and length on the product sieving and hydraulic membrane resistance (defined in section 2.3) in TFF perfusion cell culture. To measure the full fouling profile of different hollow fiber filters, initial experiments probing pore size, ID, and fiber length were conducted at high permeate flux until product sieving dropped below 10%. Then, the filters that maintained the highest product sieving were assessed at a flux that can be implemented at manufacturing scale. This work is the first, to our knowledge, to make direct comparisons of the filter performance of various manufacturers within the same studies and to directly probe the effect of the ID. The results corroborate other

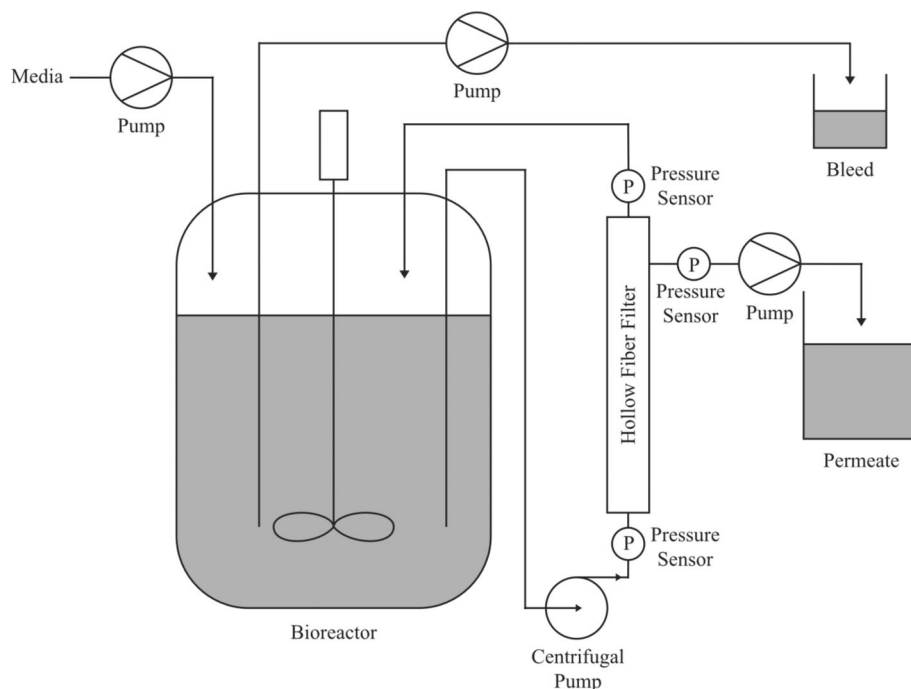
papers that have shown that at manufacturing relevant fluxes, 0.2 μm pore size filters maintain higher product sieving than 0.65 μm pore size filters, but also experience higher hydraulic membrane resistances.^{6–9} The results reveal that filters with larger IDs maintain higher product sieving and lower hydraulic membrane resistance while the shortest hollow fibers also maintained the highest product sieving, but not necessarily the lowest hydraulic membrane resistance. These insights into the contribution of geometric properties to the performance of hollow fiber filters can advise the design of future technologies that will help deliver the promises of next generation manufacturing.

2 | MATERIALS AND METHODS

2.1 | Perfusion cell culture

Experiments were performed using a proprietary Chinese Hamster Ovary (CHO) derived cell line that expressed a proprietary recombinant monoclonal antibody using in-house media. EX-CELL[®] Antifoam was added as needed (Sigma Aldrich, St. Louis, MO). All experiments were conducted in 3 L glass bioreactors (Chemglass Life Sciences, Vineland, NJ) for both the N-1 stage and production. N-1 and production bioreactors were controlled using DASGIP systems (Eppendorf, Hamburg, Germany). The perfusion set up consisted of a hollow fiber filter in a TFF configuration and a magnetically levitated centrifugal pump (Levitronix, Zurich, Switzerland) for recirculation. A schematic diagram of the setup is depicted in Figure 1. PendoTECH[®] single-use pressure sensors (Cole-Parmer, Vernon Hills, IL) were attached at the feed inlet, retentate outlet, and permeate outlet. Perfusion was maintained at a 1.2 vessel volumes per day (VVD) exchange rate in the N-1 stage starting on day 2 through day 6. Temperature was maintained at 37 °C. The pH was maintained below 7.15 by sparging CO₂. Dissolved oxygen was maintained at 50% of air saturation. The N-1 was seeded at 1e6 cells/mL and transferred to production on day 6. Production was inoculated at a 1:4 inoculum to working volume ratio and perfusion was maintained at a 1.2 VVD exchange rate starting on day 0 through day 14. The VCD was maintained at 90e6 cells/mL through a bleed. When the bleed was started, the permeate flux was lowered to maintain a 1.2 VVD exchange rate. Temperature was maintained at 35.5 °C. pH was maintained below 7.10 by sparging CO₂. Dissolved oxygen was maintained at 50% of air saturation.

The hollow fiber filter used in the N-1 stage for all experiments was S04-P20U-10-N (Repligen, Waltham, MA). The hollow fiber filters used for production are described in detail in Table 1 and include UMP-1047R, UJP-1047R, UMP-0047R, UJP-0047R, UMP-153 (Asahi Kasei, Tokyo, Japan), AU92010EXP12, AU96510EXP12, AU92020EXP12 (Sartorius, Göttingen, Germany), T04-P20U-10-N, and S04-P20U-10-N (Repligen, Waltham, MA). For membranes with areas $\leq 200\text{ cm}^2$, permeate flux was set at 5.5 LMH and working volume was adjusted to maintain the 1.2 VVD exchange rate. For membranes with areas $\geq 800\text{ cm}^2$, permeate flux was set at 1.0 LMH and working volume was adjusted to maintain the 1.2 VVD exchange rate. The recirculation rate was adjusted to maintain a shear rate of 600 s^{-1} , with the exception of the UMP-153 filter, which was maintained at 300 s^{-1} .

FIGURE 1 Schematic diagram of the perfusion bioreactor set up.**TABLE 1** Table detailing the hollow fiber filters tested for product sieving and hydraulic membrane resistance performance in a TFF configuration. Nominal pore size, chemistry, surface area, ID, and length are as stated by manufacturer.

Manufacturer	Part #	Pore Size (μm)	Chemistry	Surface Area (cm ²)	ID (mm)	Length (cm)	Fiber Count	Flux Tested (L/m ² h)
Asahi Kasei	UMP-0047R	0.2	PVDF	200	1.4	19.8	21	5.5
Asahi Kasei	UMP-1047R	0.2	PVDF	900	1.4	19.8	100	1.0
Asahi Kasei	UJP-0047R	0.65	PVDF	200	1.1	19.8	27	5.5
Asahi Kasei	UJP-1047R	0.65	PVDF	1000	1.1	19.8	135	1.0
Asahi Kasei	UMP-153	0.2	PVDF	800	2.6	19.8	50	1.0
Repligen	T04-P20U-10-N	0.2	PES	180	1.0	41.5	14	5.5
Repligen	S04-P20U-10-N	0.2	PES	980	1.0	41.5	75	1.0
Sartorius	AU92010EXP12	0.2	mPES	155	1.0	27.3	18	5.5
Sartorius	AU92020EXP12	0.2	mPES	155	2.0	27.3	9	5.5
Sartorius	AU96510EXP12	0.65	mPES	155	1.0	27.3	18	5.5

2.2 | Analytical methods

VCD and viability were measured by a Vi-CELL XR (Beckman Coulter Life Sciences, Indianapolis, IN). Protein titer was measured using high-performance liquid chromatography (Agilent Technologies, Santa Clara, CA) with a 4.6 × 50 mm POROS™ Protein A column (Thermo Fisher Scientific, Waltham, MA). pH was measured by a RAPIDPoint® 500e blood gas analyzer (Siemens, Malvern, PA).

2.3 | Calculations

TMP was calculated as follows:

$$TMP = \frac{P_f + P_r}{2} - P_p \quad (1)$$

where P_f is the feed inlet pressure, P_r is the retentate outlet pressure, and P_p is the permeate outlet pressure.

Product sieving was calculated as follows:

$$Product\ Sieving\ (\%) = \frac{Permeate\ Titer\ \left(\frac{g}{L}\right)}{Retentate\ Titer\ \left(\frac{g}{L}\right)} \times 100 \quad (2)$$

The wall shear rate (γ) was calculated as follows:

$$\gamma = \frac{4Q}{\pi r_L^3} \quad (3)$$

where Q is the volumetric flow rate and r_L is the inner (lumen) radius.

Assuming cylindrical pores, the hydraulic membrane resistance (R_t) was calculated based on the Hagen-Poiseuille equation as follows:

$$R_t = \frac{TMP}{\mu \times J} \quad (4)$$

where J is the permeate flux, TMP is the transmembrane pressure, and μ is the viscosity assumed to be the viscosity of water at 37 °C (0.69 cP). Based on the Hagen-Poiseuille equation, the hydraulic membrane resistance represents:

$$R_t = \frac{8\delta_m}{r_p^2 \epsilon} \quad (5)$$

where δ_m is the membrane thickness, ϵ is the membrane porosity, r_p is the pore radius.

The filter yield was calculated as follows:

$$\text{Filter Yield} = \frac{\text{Total Protein in Permeate (g)}}{\text{Total Protein in Permeate (g)} + \text{Remaining Protein in Bioreactor (g)}} \quad (6)$$

Note that this parameter considers only the protein that can pass through the filter and is not the overall process yield, which would include the protein lost through the bleed in the denominator.

3 | RESULTS AND DISCUSSION

3.1 | Effect of pore size

The first experiment probed the effect of pore size on filter performance. To evaluate the filters up to extensive fouling, the experiment was performed at a high permeate flux (5.5 LMH) until product sieving dropped below 10%. Four filters were compared in duplicate, including two Sartorius filters where the sole difference is a 0.2 μm pore size versus a 0.65 μm pore size (AU92010EXP12 and AU96510EXP12) and two Asahi Kasei filters where the sole differences are a 0.2 μm pore size and 1.4 mm lumen ID versus a 0.65 μm pore size and a 1.1 mm lumen ID (UMP-0047R and UJP-0047R). More details of the filter specifications can be found in Table 1. There were negligible differences in VCD and viability trends for all conditions, as seen in Figure 2a,b, respectively. The bleed was started on day 4 to maintain 90e6 cells/mL. To maintain a constant media exchange rate, permeate flux was reduced in line with the bleed, as shown in Figure 2c. Several filters began to foul significantly from day 6 onwards, resulting in a drop in permeate flux, limited by the ability of the pump to maintain the setpoint permeate flow rate as the membrane resistance increased and permeate pressure became negative. No adjustments were made to maintain the exchange rate as the permeate flux deviated from the setpoint. It should be noted that a lower permeate flux reduces the rate of fouling,⁶ however relative performance of the filters is still clear.

Product sieving over elapsed culture time and end-of-run filter yields are shown in Figure 2d,e, respectively. The Asahi Kasei 0.2 and 0.65 μm filters ended with yields of 56.2% and 47.1%, and 60.0%

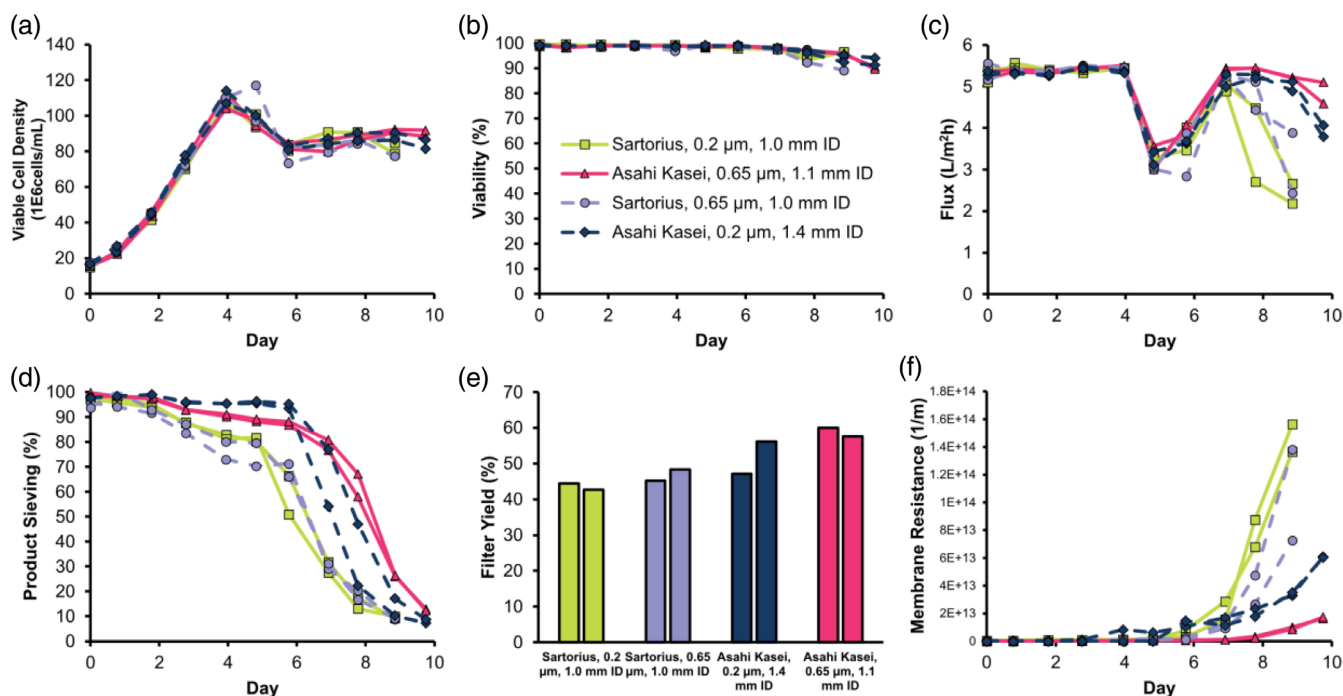


FIGURE 2 Perfusion cell culture performance for Sartorius AU92010EXP12 (light green), Sartorius AU96510EXP12 (purple), Asahi Kasei UMP-0047R (navy), and Asahi Kasei UJP-0047R (pink), in a TFF configuration: (a) viable cell density, (b) viability, (c) permeate flux, (d) product sieving, (e) end-of-run filter yield, and (f) hydraulic membrane resistance.

and 57.6%, respectively. The Sartorius 0.2 and 0.65 μm filters ended with yields of 44.5% and 42.7%, and 48.3% and 45.2%, respectively. Overall, the Asahi Kasei filters maintained higher product sieving than the Sartorius filters. Between the two Sartorius filters, there did not appear to be a difference in product sieving profiles with different pore sizes. Between the Asahi Kasei filters, the 0.2 μm filters maintained higher product sieving up to day 6, however, fouled faster than the 0.65 μm filters, leading to a lower overall yield. Generally, literature has seen no improvement or worse performance in product sieving when using 0.45–0.65 μm pore filters compared to 0.2 μm pore filters.^{6–8} However, these studies assessed the filters at lower fluxes than tested here, closer to manufacturing relevant fluxes (1 and 2 LMH).

Figure 2f shows the hydraulic membrane resistance over elapsed culture time. Both Sartorius filters ended with higher hydraulic membrane resistances than the Asahi Kasei filters. Furthermore, both 0.2 μm filters ended with higher hydraulic membrane resistances than their equivalent 0.65 μm filter counterpart. This result aligns with previous papers that

have found that 0.2 μm filters experience larger TMP increases compared to 0.40 or 0.65 μm filters.⁹ This suggests larger pore sizes can be beneficial for maintaining lower TMPs below process limitations.

3.2 | Effect of lumen ID and fiber length

The second experiment probed the effect of lumen ID and fiber length. To evaluate the filters up to extensive fouling, this experiment was also performed at a high permeate flux (5.5 LMH) until product sieving dropped below 10%. Four filters were compared in duplicate, including Asahi Kasei UMP-0047R (pore size: 0.2 μm , lumen ID: 1.4 mm ID), Sartorius AU92010EXP12 (0.2 μm , 1.0 mm ID), Sartorius AU92020EXP12 (0.2 μm , 2.0 mm ID), and Repligen T04-P20U-10-N (0.2 μm , 1.0 mm ID), details for which can be found in Table 1. There were negligible differences in VCD and viability trends for all conditions, as seen in Figure 3a,b, respectively. The bleed was started on

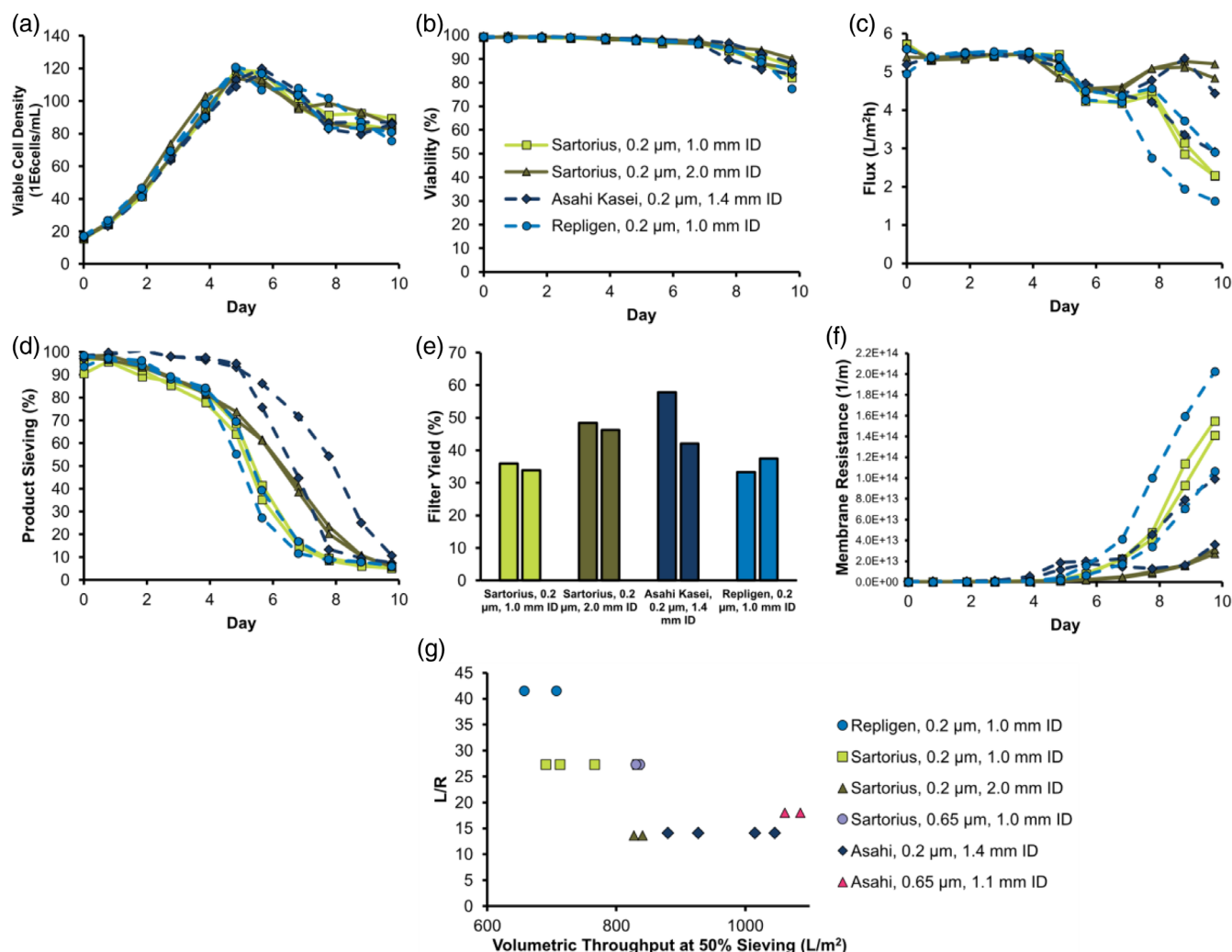


FIGURE 3 Perfusion cell culture performance for Asahi Kasei UMP-0047R (navy), Sartorius AU92010EXP12 (light green), Sartorius AU92020EXP12 (dark green), and Repligen T04-P20U-10-N (light blue), in a TFF configuration: (a) viable cell density, (b) viability, (c) permeate flux, (d) product sieving, (e) end-of-run filter yield, (f) hydraulic membrane resistance, and (g) length to radius ratio (L/R) versus volumetric throughput at 50% sieving.

day 4 to maintain 90e6 cells/mL. To maintain a constant media exchange rate, permeate flux was reduced in line with the bleed, as shown in Figure 3c. As noted for the first experiment, several filters began to foul significantly from day 6 onwards and no adjustments were made to maintain the setpoint exchange rate.

Product sieving over the elapsed culture time and end-of-run filter yields are shown in Figure 3d,e, respectively. On average, the Asahi Kasei (0.2 µm, 1.4 mm ID) filters maintained the highest product sieving and ended with yields of 57.9% and 42.0%. As there were negligible differences in cell culture performance between these two duplicates, it is likely that the variability in this duplicate is due to an exacerbation of normal process variability due to the high permeate flux. Despite the variability, the relative performance of the filters is still clear. The other filters had comparable sieving up to day 4, however from day 4 onwards, the Sartorius (0.2 µm, 2.0 mm ID) filters maintained higher product sieving, ending with yields of 48.4% and 46.3%. The Sartorius (0.2 µm, 1.0 mm ID) and Repligen (0.2 µm, 1.0 mm ID) filters had the sharpest product sieving decays. The end-of-run yields were 35.9% and 33.8% for Sartorius (0.2 µm, 1.0 mm ID) and 37.5% and 33.3% for Repligen (0.2 µm, 1.0 mm ID).

Figure 3f shows the hydraulic membrane resistance over the elapsed culture time. The Sartorius, (0.2 µm, 1.0 mm ID) and Repligen (0.2 µm, 1.0 mm ID) filters showed similar profiles on average and ended with the highest hydraulic membrane resistances of all conditions, followed by the Asahi Kasei (0.2 µm, 1.4 mm ID) filters. Notably, the Sartorius (0.2 µm, 2.0 mm ID) filters maintained the lowest hydraulic membrane resistance.

Comparing the Sartorius filters, the larger lumen ID appeared to improve product sieving and hydraulic membrane resistance. We hypothesize that the reduction in axial pressure drop due to the larger lumen ID reduces the pressure gradient along the length of the fiber leading to a reduction in the fouling rate. In this process, prior to significant fouling, the permeate pressure is higher than the retentate outlet pressure indicating that the permeate is back-filtering across the membrane near the outlet. A reduced pressure gradient would reduce the extent of this back-filtration at the same permeate flow rate. Similarly, however, if the permeate pressure were lower than the outlet pressure, reducing the pressure gradient would still lead to a more axially uniform flux profile, and therefore more efficient use of the membrane area. This aligns with the findings of Binabaji et al., where it was shown that shorter hollow fibers experience less back-filtration due to the reduced pressure drop, as described by the Hagen-Poiseuille equation (defined below).¹⁴

$$\Delta P = \frac{8\mu LQ}{\pi R^4} \quad (7)$$

Based on the Hagen-Poiseuille equation, the dynamic viscosity (μ), length of the hollow fibers (L), volumetric axial cross flow rate (Q), and inner radius (R) contribute to the axial pressure drop, in turn contributing to the amount of back-filtration causing filter fouling. In this paper, all hollow fiber filters were assessed at the same shear rate, the equation for which is defined below:

$$\gamma = \frac{4Q}{\pi R^3} \quad (8)$$

Combining equations [7] and [8] to assess the pressure drop at constant shear rate, we obtain:

$$\Delta P = \frac{2\mu L\gamma}{R} \quad (9)$$

Importantly,

$$\Delta P \propto \frac{L}{R} \quad (10)$$

This relationship suggests shorter fiber lengths and larger lumen IDs should proportionally improve back-filtration during TFF operation. The relative performance of the Sartorius filters supports this as the larger lumen ID showed an improvement in product sieving and hydraulic membrane resistance when shear rate was held constant. The highest product sieving profile was also observed in the Asahi Kasei (0.2 µm, 1.4 mm ID) filters, which have the shortest length. Notably however, comparing the Sartorius (0.2 µm, 1.0 mm ID) and Repligen (0.2 µm, 1.0 mm ID) filters, the shorter length of the Sartorius filter did not appear to improve the product sieving performance or hydraulic membrane resistance. However, differences in manufacturer or membrane chemistry could be confounding factors affecting filter performance. Specifically, when studying filters of different membrane chemistries, Nikolay et al. found variations in pore size distributions, surface roughness, and porosity when measuring by capillary flow porometry and scanning electron microscopy (SEM) imaging.¹⁵ Figure 3g shows the length to radius ratio (L/R) of filters from both experiments run at 5.5 LMH graphed against their total volumetric throughput once 50% product sieving is reached. As expected, based on the Hagen-Poiseuille equation, there is a general negative correlation between L/R and volumetric throughput.

Because the Asahi Kasei filters maintained the highest product sieving and low hydraulic membrane resistance, filters from this manufacturer were further studied at a setpoint flux of 1.0 LMH.

3.3 | Implementation in manufacturing

In manufacturing, large TMP increases due to fouling can lead to unstable process operation and process failure. The purpose of evaluating filters at 1.0 LMH was to understand their relative performance at a flux setpoint that will maintain operable TMPs for a full-length process. Three filters manufactured by Asahi Kasei were evaluated: the UMP-1047R (pore size: 0.2 µm, lumen ID: 1.4 mm), UJP-1047R (0.65 µm, 1.1 mm ID), and UMP-153 (0.2 µm, 2.6 mm ID). The Repligen S04-P20U-10-N (0.2 µm, 1.0 mm ID) filter was also included in the experiment as an additional reference point to the previous experiment. More details of the filter specifications can be found in Table 1. A shear rate of 600 s⁻¹ through the fiber was kept constant for all conditions, with the exception of the UMP-153 filter, which was run

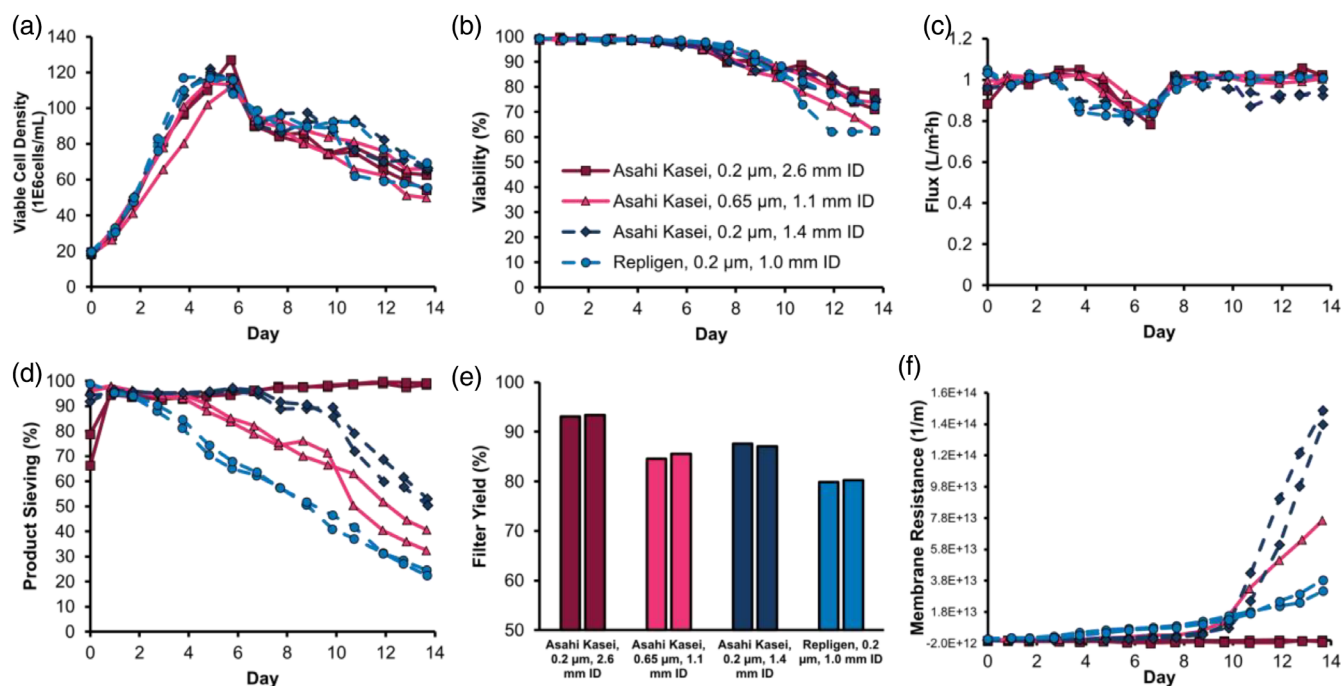


FIGURE 4 Perfusion cell culture performance for Asahi Kasei UMP-1047R (navy), Asahi Kasei UJP-1047R (pink), Asahi Kasei UMP-153 (magenta), and Repligen S04-P20U-10-N (light blue) in a TFF configuration: (a) viable cell density, (b) viability, (c) permeate flux, (d) product sieving, (e) end-of-run filter yield, and (f) hydraulic membrane resistance.

at a shear rate of 300 s^{-1} . This is because a shear rate of 600 s^{-1} through the filter required very high rotational speeds from the Levitronix centrifugal pump and there was a concern that the shear caused by the pump would adversely impact the cell culture.

There were negligible differences in VCD and viability trends for all conditions, as seen in Figure 4a,b, respectively. The bleed was started on day 4 to maintain 90×10^6 cells/mL. To maintain a constant media exchange rate, permeate flux was reduced in line with the bleed as shown in Figure 4c. Product sieving against elapsed culture time and end-of-run filter yields are shown in Figure 4d,e, respectively. The Asahi Kasei (0.2 µm, 2.6 mm ID) filters maintained product sieving above 98% for the entire duration of the cell culture, ending with yields of 93.4% and 93.0%. The Asahi Kasei (0.2 µm, 1.4 mm ID) filters showed some product sieving decay ending with yields of 87.6% and 87.1%. This was slightly higher than the Asahi Kasei (0.65 µm, 1.1 mm ID) filters, which ended at yields of 85.6% and 84.5%. The Repligen (0.2 µm, 1.0 mm ID) filters had the lowest product sieving, ending at yields of 80.2% and 79.8%. Overall, the filters showed a similar negative correlation between L/R and volumetric throughput at 50% sieving as that seen in Figure 3g (Supplementary Figure 1); indeed, the Asahi Kasei (0.2 µm, 2.6 mm ID) filters, with the lowest L/R at 7.6, maintained sieving >98% throughout the duration of the experiment.

Figure 4f shows the hydraulic membrane resistance against elapsed culture time. The Asahi Kasei (0.2 µm, 1.4 mm ID) filters ended with the highest hydraulic membrane resistance, followed by the Asahi Kasei (0.65 µm, 1.1 mm ID) filters, the Repligen (0.2 µm, 1.0 mm ID) filters, and then the Asahi Kasei (0.2 µm, 2.6 mm ID) filters, which maintained the lowest hydraulic membrane resistance.

The product sieving profiles of the Asahi Kasei filters in this study align with the findings of Pinto et al., where the 0.65 µm Asahi Kasei

filter performed worse than the 0.2 µm Asahi Kasei filter in product sieving and TMP at manufacturing relevant fluxes (1 and 2 LMH). They attributed the minimal fouling of the 0.2 µm filter to wide-surface conical shaped pores measured by SEM imaging that were not seen in the 0.65 µm filter.⁶ Although the Asahi Kasei (0.2 µm, 2.6 mm ID) filters maintained the highest product sieving relative to the other filters tested, several papers have demonstrated that lower shear along the fiber leads to less filter fouling measured either by TMP increases or product sieving decline.^{6,9,11,16} This is in line with the hypothesis of the influence of axial pressure drop on the back-filtration: lower shear rates reduce the axial pressure drop, reducing the back filtration, and should therefore improve product sieving. Considering the Hagen-Poiseuille equation [7], half the shear rate should reduce the axial pressure drop by a factor of 2, whereas an increase in lumen diameter from 1.4 to 2.6 mm should reduce the axial pressure drop by a factor of 1.86. Therefore, in this study, the Asahi Kasei (0.2 µm, 2.6 mm ID) condition experiences an additional benefit from operating at a lower shear rate.

4 | CONCLUSIONS

This work evaluated the product sieving and hydraulic membrane resistance performance of various hollow fiber filter geometries. The work presented here is the first, to our knowledge, to evaluate filters of various manufacturers within the same studies and to make comparisons of different IDs. It was found that larger IDs improve the product sieving and hydraulic membrane resistance, hypothesized to be because of a reduction in back-filtration. Larger pore sizes were found to reduce the hydraulic membrane resistance but perform no

different or worse in regards to product sieving. Filters manufactured by Asahi Kasei maintained the highest product sieving and lowest hydraulic membrane resistance overall, likely because the shorter length reduces the back-filtration. We have additionally provided evidence that a combination of a shorter length hollow fiber, larger lumen ID, and lower shear rate can result in a significant reduction in fouling given the same starting cell culture material. These results not only provide further insights into the fouling mechanism of hollow fiber membranes but can inform the design of filter technologies adopted to perfusion cell culture.

AUTHOR CONTRIBUTIONS

Dominique WuDunn: Methodology; conceptualization; investigation; writing – original draft; writing – review and editing; visualization; formal analysis; data curation. **Andrea Squeri:** Supervision; conceptualization; writing – review and editing; methodology. **Jimmy Vu:** Conceptualization; writing – review and editing; methodology. **Ashna Dhingra:** Formal analysis. **Jon L. Coffman:** Supervision; writing – review and editing; conceptualization. **Ken Lee:** Supervision; writing – review and editing; conceptualization.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

PEER REVIEW

The peer review history for this article is available at <https://www.webofscience.com/api/gateway/wos/peer-review/10.1002/btpr.3440>.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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REFERENCES

- Clincke MF, Mölleryd C, Samani PK, et al. Very high density of Chinese hamster ovary cells in perfusion by alternating tangential flow or tangential flow filtration in WAVE bioreactor™—part II: applications for antibody production and cryopreservation. *Biotechnol Prog*. 2013;29(3):768–777. doi:10.1002/btpr.1703
- Wang S, Godfrey S, Ravikrishnan J, Lin H, Vogel J, Coffman J. Shear contributions to cell culture performance and product recovery in ATF and TFF perfusion systems. *J Biotechnol*. 2017;246:52–60. doi:10.1016/j.jbiotec.2017.01.020
- Karst DJ, Serra E, Villiger TK, Soos M, Morbidelli M. Characterization and comparison of ATF and TFF in stirred bioreactors for continuous mammalian cell culture processes. *Biochem Eng J*. 2016;110:17–26. doi:10.1016/j.bej.2016.02.003
- Blaschczok K, Kaiser SC, Löffelholz C, et al. Investigations on mechanical stress caused to CHO suspension cells by standard and single-use pumps. *Chemie Ingenieur Technik*. 2013;85(1–2):144–152. doi:10.1002/cite.201200135
- Coffman J, Brower M, Connell-Crowley L, et al. A common framework for integrated and continuous biomanufacturing. *Biotechnol Bioeng*. 2021;118(4):1735–1749. doi:10.1002/bit.27690
- Pinto NDS, Brower M. Wide-surface pore microfiltration membrane drastically improves sieving decay in TFF-based perfusion cell culture and streamline chromatography integration for continuous bioprocessing. *Biotechnol Bioeng*. 2020;117(11):3336–3344. doi:10.1002/bit.27504
- Pinto NDS, Napoli WN, Brower M. Impact of micro and macroporous TFF membranes on product sieving and chromatography loading for perfusion cell culture. *Biotechnol Bioeng*. 2020;117(1):117–124. doi:10.1002/bit.27192
- Wang SB, Godfrey S, Radonqi F, Lin H, Coffman J. Larger pore size hollow fiber membranes as a solution to the product retention issue in filtration-based perfusion bioreactors. *Biotechnol J*. 2019;14(2):1800137. doi:10.1002/biot.201800137
- Stressmann M, Moresoli C. Effect of pore size, shear rate, and harvest time during the constant permeate flux microfiltration of CHO cell culture supernatant. *Biotechnol Prog*. 2008;24(4):890–897. doi:10.1002/btpr.4
- Xu S, Jiang R, Chen Y, Wang F, Chen H. Impact of Pluronic® F68 on hollow fiber filter-based perfusion culture performance. *Bioprocess Biosyst Eng*. 2017;40(9):1317–1326. doi:10.1007/s00449-017-1790-2
- Radonqi F, Zhang H, Bardliving CL, Shamlou P, Coffman J. Computational fluid dynamic modeling of alternating tangential flow filtration for perfusion cell culture. *Biotechnol Bioeng*. 2018;115(11):2751–2759. doi:10.1002/bit.26813
- Bolton G, LaCasse D, Kuriyel R. Combined models of membrane fouling: development and application to microfiltration and ultrafiltration of biological fluids. *J Memb Sci*. 2006;277(1):75–84. doi:10.1016/j.memsci.2004.12.053
- Binabaji E, Ma J, Rao S, Zydney AL. Theoretical analysis of the ultrafiltration behavior of highly concentrated protein solutions. *J Memb Sci*. 2015;494:216–223. doi:10.1016/j.memsci.2015.07.068
- Binabaji E, Ma J, Rao S, Zydney AL. Ultrafiltration of highly concentrated antibody solutions: experiments and modeling for the effects of module and buffer conditions. *Biotechnol Prog*. 2016;32(3):692–701. doi:10.1002/btpr.2252
- Nikolay A, de Grooth J, Genzel Y, Wood JA, Reichl U. Virus harvesting in perfusion culture: choosing the right type of hollow fiber membrane. *Biotechnol Bioeng*. 2020;117(10):3040–3052. doi:10.1002/bit.27470
- Su Y, Wei Z, Miao Y, et al. Optimized process operations reduce product retention and column clogging in ATF-based perfusion cell cultures. *Appl Microbiol Biotechnol*. 2021;105(24):9125–9136. doi:10.1007/s00253-021-11662-8

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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